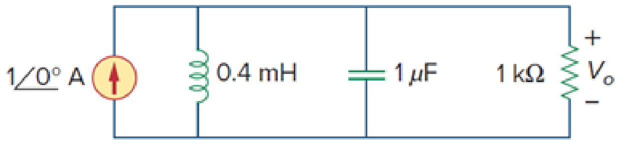
Chapter 14, Problem 16PP

Problem

Consider the network in Fig. 14.57. Use *PSpice* to obtain the Bode plots for *Vo* over a frequency from 1 to 100 kHz using 20 points per decade.

Figure 14.57:



Step-by-step solution

Step 1 of 7

Step 1:

We use schematics to create the circuit, and double click on each component and enter the given values as shown in below circuit.

Here, we use VAC voltage source from the source library and double click on VAC voltage source and set DC = 0V, ACMAG = 1V and ACPHASE = 00

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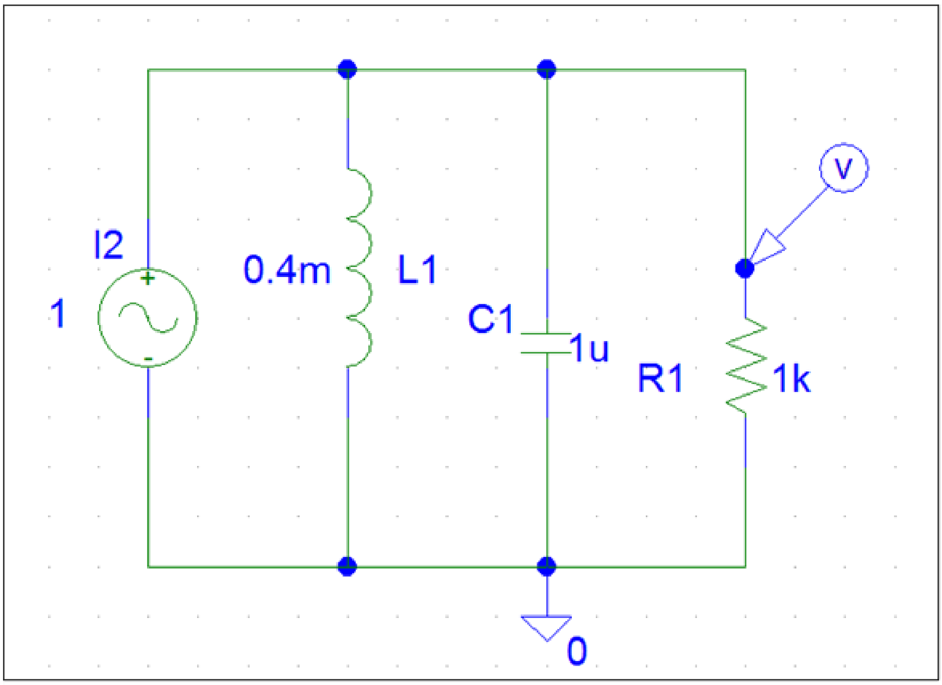


Step 3 of 7

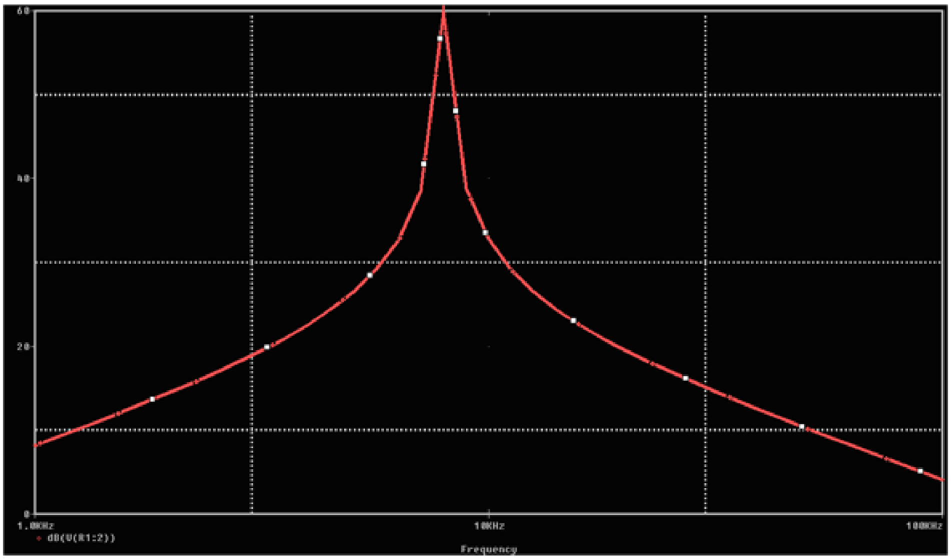
Step 2:

In the simulation analysis, select AC sweep, and enter the starting frequency $=1\ \mathrm{kHz}$ end frequency $=100\ \mathrm{kHz}$ and total points/decade $=20$

Step 3:

Go to ‘Analysis’ menu open ‘simulate’ (or press ‘F- 11’)

The resultant magnitude plot is



Step 4:

To obtain the phase plot we select **trace / add** in PSpice A/D demo menu and type VP(R1:2) in the trance command box. The resultant bode phase plot is